202606UECM3473OE1a

Start again

Review of preview

Started on	Tuesday, 30 June 2026, 07:53 AM
Completed on	Tuesday, 30 June 2026, 07:54 AM
Time taken	11 secs
Grade	**0** out of a maximum of 10 (**0**%)

1

Marks: 1

Let X and Y have joint probability density function(pdf) $f(x,y) = ce^{-6y}$, $0 < x < y < \infty$ and zero otherwise. Find the mean of the conditional distribution of $Y|X=1.35$. ______

Answer:

✗

Make comment or override grade

Incorrect

Correct answer: 1.516667

Marks for this submission: 0/1.

2 Marks: 1

Let X_1 and X_2 be independent random variables. X_1 follows a gamma distribution with parameters $\alpha = 3$ and $\beta = 1/8$, whereas X_2 follows an exponential distribution with mean 1/4. Find $E[X_1|X_1 + X_2 = 5]$. ______

Answer:

Make comment or override grade

Incorrect
Correct answer: 0.75

Marks for this submission: 0/1.

3 Marks: 1

On an auto collision coverage, there are 3 classes of policyholders, A , B and C. 39% of drivers are in class A, 25% of the drivers are in class B and 36% in class C. The distribution of losses for the drivers are:

Class	Losses		
	1000	5000	10000
	Probability		
A	0.48	0.26	0.26
B	0.53	0.32	0.15
C	0.54	0.25	0.21

A claim is submitted by a randomly selected driver. Calculate the standard deviation of the size of the claim. ______

Answer:

Make comment or override grade

Incorrect
Correct answer: 3552.469675

Marks for this submission: 0/1.

4 Marks: 1

Given a value of $(\Theta = \theta)$, the random variable X follows a Gamma distribution with probability density function

$$f(x) = \theta^2 x e^{-\theta x}.$$

Θ has a uniform distribution on the interval (4, 9). Determine $S_X(0.64)$ for the unconditional distribution. ______

Answer:

Make comment or override grade

Incorrect
Correct answer: 0.102518

Marks for this submission: 0/1.

5
Marks: 1

Annual claim counts per risk are binomial with parameter m = 3 and Q. Q varies by risk uniformly on (0.33, 0.83). For a risk selected at random, determine the probability of at most one claims. ______

Answer:

Make comment or override grade

Incorrect
Correct answer: 0.39103

Marks for this submission: 0/1.

6
Marks: 1

You are given:

- Conditional on q, the random variables X_1, X_2,..., X_m, are independent and follow a Bernoulli distribution with parameter q.
- $S_k = X_1 + X_2 + \ldots + X_k$.
- The distribution of q is Beta(a=2, b = 2, θ = 12).

Determine the variance of the marginal distribution of S_{113}. ______

Answer:

Make comment or override grade

Incorrect
Correct answer: 87733.2

Marks for this submission: 0/1.

7
Marks: 2

The size of loss ,X has mean 4λ and variance $7\lambda^2$. Λ has the following density function:

$$f(\lambda) = 5(1250)^5/(\lambda+1250)^6.$$

Calculate the variance of the loss. ______

Answer:

Make comment or override grade

Incorrect
Correct answer: 4427083.333333

Marks for this submission: 0/2.

8
Marks: 2

The annual number of accidents for an individual driver has a Poisson distribution with mean λ. The mean, Λ, of a heterogeneous population of drivers have a gamma distribution with mean 0.22 and variance 0.0242. Calculate the probability that a driver selected at random from the population will have 2 or more accidents in one year. ______

Answer:

Make comment or override grade

Incorrect
Correct answer: 0.0275

Marks for this submission: 0/2.